

Tingyu Wu

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RESEARCH INTERESTS

Stochastic optimization and decision-making under uncertainty, with a focus on distributionally robust optimization and learning theory

EDUCATION

University of Illinois Urbana-Champaign Aug. 2023 – Dec. 2026 (Expected)
B.S. Mathematics, Concentration in Data Optimization, Minor in Statistics

SKILLS

Languages: Python, R, $\text{\LaTeX}$, Java, C++
Tools: Git/GitHub, Docker
Libraries: NumPy, pandas, Matplotlib, PyTorch

RESEARCH EXPERIENCE

- Independent Research** Jul. 2025 – Present
Prof. Yifan Hu, Department of Statistics, Rutgers University *Remote*
- Establish high-probability solution-set convergence guarantees for continuous sample-average approximation and nonconvex-concave stochastic minimax optimization, deriving corresponding sample-complexity bounds.
 - Strengthen the convergence analysis of a regularized minimax method for causal invariance learning, providing a complete proof and improving the existing convergence rate.
 - Investigate causal identification in BLP demand models through invariance across environments, exploring gradient invariance as an identification condition and its connections to instrumental-variable and GMM formulations.
- Undergraduate Research Scholar** Sept. 2025 – Dec. 2025
Prof. Felix Leditzky, Illinois Mathematics Lab (IML), University of Illinois Urbana-Champaign *Champaign, IL*
- Benchmarked classical optimization methods for entropy-based problems in quantum information theory, focusing on convergence and numerical stability.
 - Analyzed algorithmic performance via performance and data profiles and implemented optimization routines for numerical evaluation.
 - Awarded the Nancy D. Anderson Undergraduate Research Award** for this project; [manuscript](#).

PROJECTS

- Agent-based CBT Intervention** | *LLM, LangChain, JSON, Agent, Document AI, Python* May 2025 – Jul. 2025
- Built an intelligent agent to provide initial mental health screening and personalized CBT suggestions based on student questionnaire scores.
 - Developed a complete pipeline including MHT score parsing, JSON conversion, and interactive agent responses.
- LaTeX-OCR** | *$\text{\LaTeX}$, LLM, Im2LaTeX, Document AI, Python* Nov. 2023 – May 2024
- Generated a robust dataset to train the LaTeX-OCR model using the 7B Zephyr LLM.
 - Analyzed open-source LaTeX-OCR models, identified key distinctions, and implemented optimization strategies.
- Hyper-parameter Optimization** | *Neural Network, Genetic Algorithm, Transfer Learning* May 2022 – Sept. 2022
- Collaborated on a group paper, Hyper-parameter Optimization Based on GA & Transfer Learning.
 - Compared existing hyper-parameter optimization methods and designed a novel algorithm leveraging genetic search and transfer learning.
 - Implemented experiments on the MNIST dataset and demonstrated improvement.